STAT 7650: Computational Statistics

7. Integration and Simulation

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Outline

1. Numerical integration
2. Exact simulation
3. Approximate simulation
4. Variance reduction techniques

Integration

Let X be a random variable with density $f(x)$. For a function $h(x)$,

$$E[h(X)] = \int h(x)f(x)dx$$

The examples include

- moments, such as mean, variance, ...
- probability, $h(x) = I_A(x)$ for a set A

Two different approaches

- numerical integration
- Monte Carlo integration. Draw a sample $x_1, \ldots, x_n$ from $f(x)$,

$$E[h(X)] \approx \frac{1}{n}\sum_{i=1}^{n} h(x_i)$$

Numerical Integration by Riemann Sum

Evaluate a one-dimensional integral for a univariate function $f(x)$,

$$\int_a^b f(x)dx$$

Partition $[a, b]$ into n subintervals, $[x_i, x_{i+1}]$ for $i = 0, \ldots, n-1$,

$$a = x_0 < x_1 < x_2 < \cdots < x_{n-1} < x_n = b$$

Approximate the integral by the Riemann sum

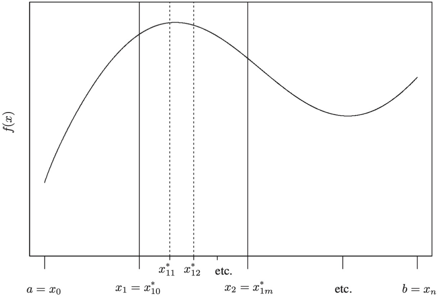
$$\int_a^b f(x)dx = \sum_{i=0}^{n-1} \int_{x_i}^{x_{i+1}} f(x)dx \approx \sum_{i=0}^{n-1} f(t_i)(x_{i+1} - x_i)$$

where $t_i \in [x_i, x_{i+1}]$. Usually choose $t_i = x_i$ for convenience.

Numerical Integration

In more general, partition $[a, b]$ into n subintervals, $[x_i, x_{i+1}]$ for $i = 0, \ldots, n-1$ with $x_0 = a$ and $x_n = b$. Within each subinterval, insert $m+1$ nodes x_{ij}^* for $j = 1, \ldots, m$. Then

$$\int_a^b f(x)dx = \sum_{i=0}^{n-1} \int_{x_i}^{x_{i+1}} f(x)dx \approx \sum_{i=0}^{n-1} \sum_{j=0}^{m} A_{ij} f(x_{ij}^*)$$

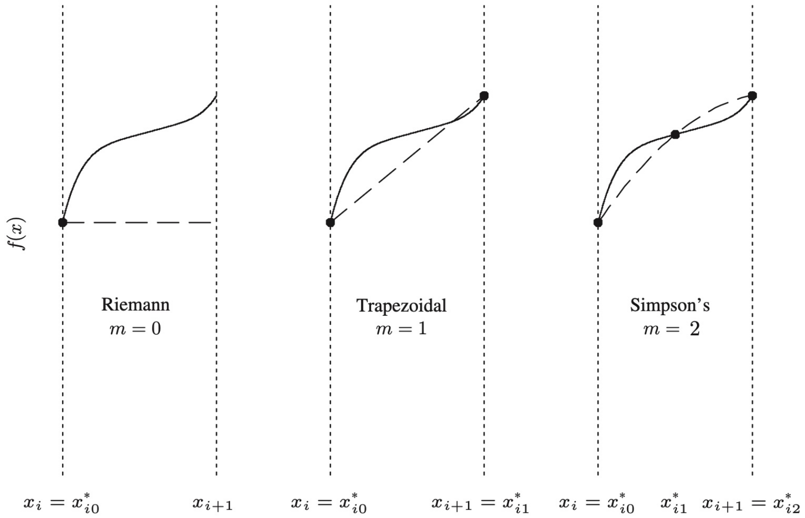


Newton-Cotes Quadrature

In general, the subintervals may have different lengths and the numbers of nodes may vary within each subintervals.

For the Newton-Cotes rule,

- the nodes are equally spaced in $[x_i, x_{i+1}]$
- the same number of nodes is used in each subintevral
- replace $f(x)$ with a polynomial approximation on each subinterval
    - Riemann rule: $m = 0$, $x_{i0}^* = x_i$, $A_{i0} = x_{i+1} - x_{i0}^*$
    - Trapezoidal rule: $m = 1$
    - Simpson's rule: $m = 2$



$f(x)$
Riemann
$m = 0$
Trapezoidal
$m = 1$
Simpson's
$m = 2$
$x_i = x^*_{i0}$
x_{i+1}
$x_i = x^*_{i0}$
$x_{i+1} = x^*_{i1}$
$x_i = x^*_{i0}$
x^*_{i1}
$x_{i+1} = x^*_{i2}$

R Function

The `integrate()` function can compute the integration of a univariate function.

```
integrate(f, lower, upper, ...)
```

Compute the first two moments of $N(2, 3^2)$.

```
integrate(function(x) { x * dnorm(x, 2, 3) }, -Inf, Inf)
integrate(function(x) { x^2 * dnorm(x, 2, 3) }, -Inf, Inf)
```

Monte Carlo Integration

Let the density of X be $f(x)$. Let $X_1, \ldots, X_n$ be an iid sample. By the strong law of large numbers, as $n \to \infty$,

$$\hat{\mu}_{\mathsf{MC}} = \frac{1}{n} \sum_{i=1}^{n} h(X_i) \longrightarrow \mu = E[h(X)] = \int h(x) f(x) dx$$

and when $\sigma^2 = var[h(X)] = E[h(X) - \mu]^2$ exists,

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} [h(X_i) - \hat{\mu}_{\mathsf{MC}}]^2 \longrightarrow \sigma^2$$

Hence a $100(1-\alpha)\%$ confidence interval of μ is

$$\hat{\mu}_{\mathsf{MC}} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$$

Monte Carlo Methods for Quadrature

To evaluate the following integral

$$I = \int_D g(x)dx,$$

suppose X is a random variable with density $f(x)$ with support on D. Then $h(x) = g(x)/f(x)$ and

$$I = \int_D h(x)f(x)dx = E[h(X)].$$

1. generate a sequence of random numbers $x_1, \ldots, x_n$ from $f(x)$
2. approximate I by

$$\hat{I} = \frac{1}{n}\sum_{i=1}^{n} h(x_i).$$

Variance of Monte Carlo Estimators

The Monte Carlo estimate $\hat{I}$ is unbiased and its variance is

$$var(\hat{I}) = \frac{1}{n} E[(h(X) - I)^2].$$

An estimate for the variance is

$$\widehat{var}(\hat{I}) = \frac{1}{n(n-1)} \sum_{i=1}^{n} (h(x_i) - \hat{I})^2.$$

To get a better estimate of I,

- the smaller variation in $h(x) - I$
- the larger n

Example: Evaluating Probability

Randomly select three numbers independently from $U(0,1)$. What is the probability that they can be the lengths of sides in a triangle?

1. Initialization. Set $K = 10,000$ and $n = 0$.
2. For each $i = 1, \ldots, K$,
   1. randomly generate three numbers u_1, u_2, u_3 independently from $U(0,1)$.
   2. if they can form a triangle ($u_1 + u_2 > u_3$, $u_1 + u_3 > u_2$, $u_2 + u_3 > u_1$), then $n = n + 1$.
3. Output. The probability is n/K.

Solution

Let U_1, U_2, U_3 are independently Uniform(0, 1) and let $U_{(1)} \le U_{(2)} \le U_{(3)}$ are order statistics. Then we want to evaluate

$$P(U_{(1)} + U_{(2)} > U_{(3)}).$$

The joint density of $U_{(1)}$, $U_{(2)}$, $U_{(3)}$ is $f(u_1, u_2, u_3) = 6$ for $0 < u_1 \le u_2 \le u_3 < 1$. Then the desired probability is

$$\int_{u_1+u_2>u_3} 6du_1du_2du_3 = \int_0^1 \int_{u_3/2}^{u_3} \int_{u_3-u_2}^{u_2} 6du_1du_2du_3 = \frac{1}{2}.$$

Question: If I want the result from Monte Carlo to an accuracy of three decimal places (with probability 95%), what is the minimum value of K?

Example: Evaluating Moments

Let $X_1, \ldots, X_n$ are iid Uniform(0, 1) and $X_{(1)}, \ldots, X_{(n)}$ are order statistics. Let $T_k = X_{(k)} - X_{(k-1)}$.

- Calculate the mean and standard deviation of T_k.
- Do the mean and standard deviation of T_k depend on k?

1. Initialization. Set $B = 10,000$.
2. For each $i = 1, \ldots, B$,
   1. randomly generate n numbers $X_1, \ldots, X_n$ from U(0, 1)
   2. Calculate $T_{k,i} = X_{(k)} - X_{(k-1)}$
3. Output the average and standard deviation of $T_{k,i}$.

Solution

Write out the joint density of $X_{(k-1)}$ and $X_{(k)}$ and then do transformation $T = X_{(k)} - X_{(k-1)}$ and $W = X_{(k-1)}$. After integrating out w, the density of T is

$$f(t) = \int \frac{n!}{(k-2)!\,(n-k)!}[F(w)]^{k-2}[1-F(t+w)]^{n-k} f(w) f(t+w) dw$$

Notice that the values of t and w satisfy $0 \le t < 1$, $0 < w < 1$, and $0 < t + w < 1$. Then the density of T is Beta(1, n).

$$\begin{aligned} f(t) &= \frac{n!}{(k-2)!\,(n-k)!} \int_0^{1-t} w^{k-2}(1-t-w)^{n-k} dw \\ &= \frac{n!}{(k-2)!\,(n-k)!}(1-t)^{n-1} B(k-1, n-k+1) \\ &= n(1-t)^{n-1}. \end{aligned}$$

We have $E(T) = 1/(n+1)$, $var(T) = n/[(n+1)^2(n+2)]$

Monte Carlo Integration vs Quadrature Methods

- Convergence rate for Monte Carlo integration: $O(n^{-1/2})$. To reduce the error by half, the sample size need to be quadrupled.
- Convergence rate for quadrature methods: $O(n^{-2})$ or better
- Quadrature methods perform best when h is smooth, while Monte Carlo integration approach ignores smoothness.
- The order of error is $O(n^{-1/2})$ only depends on the sample size, not on dimension.
- Quadrature methods are difficult to extend to multidimensional problems. The implementation of Monte Carlo integration is less hampered by high dimensionality.
- For univariate or low-dimensional integral, numerical quadrature is usually better. For high-dimensional integral, Monte Carlo estimate is usually better.

Standard Parametric Distribution Families

The desired sampling density f is called the target distribution.

- uniform$(0, 1)$: `runif()`
  - pseudo-random numbers
  - random seed: `set.seed()`
- Other standard parametric distribution families
  - normal distribution $N(\mu, \sigma^2)$: `rnorm()`
  - exponential distribution $\exp(\lambda)$: `rexp()`
  - Gamma distribution Gamma(a, b): `rgamma()`
  - Poisson distribution Poisson(λ): `rpois()`
  - multivariate normal distribution $N(\mu, \Sigma)$: `mvrnorm()` in `MASS`
  - Wishart distribution $W(k, \Psi)$: `rWishart()`

Inverting CDF

For any continuous distribution function F, if $U \sim \text{unif}(0, 1)$, then $X = F^{-1}(U) = \inf\{x : F(x) \geq U\}$ has the CDF F.

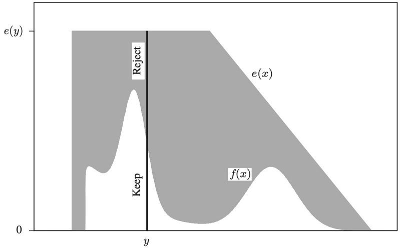
- discrete distribution, for example, binomial(n, π)
- exponential distribution exp(λ)
- If F^{-1} is not available but F is either available or easily approximated, then approximate F^{-1} by linear interpolation.
  - choose a grid $x_1, \ldots, x_m$ spanning the support of f
  - calculate $u_i = F(x_i)$ for $i = 1, \ldots, m$
  - Draw $U \sim \text{unif}(0, 1)$, If $u_i \leq U \leq u_j$, then

$$X = \frac{u_j - U}{u_j - u_i} x_i + \frac{U - u_i}{u_j - u_i} x_j$$

Rejection Sampling

Let f be the target density. Let g denote another density and $e(\cdot)$ denote an envelope such that $e(x) = g(x)/\alpha \geq f(x)$ for $\alpha \leq 1$ and all x satisfying $f(x) > 0$.

- Sample $Y \sim g$
- Sample $U \sim \text{unif}(0, 1)$
- Output $X = Y$ if $U \leq f(Y)/e(Y)$; otherwise reject Y and repeat.



It can be proved that $X \sim f(x)$. Thus, the sampling distribution is exact. The constant α is a measure of the efficiency of the algorithm, which is the expected proportion of candidates that are accepted.

Example: Gamma Distribution

Let X be a Gamma(r, 1) random variable with $r \geq 1$ and density

$$f(x) = \frac{1}{\Gamma(r)} x^{r-1} e^{-x}$$

Let $a = r - (1/3)$, $b = 1/\sqrt{9a}$, and $X = t(Y) = a(1 + bY)^3$. Density

$$f(y) = \frac{1}{\Gamma(r)} [t(y)]^{r-1} e^{-t(y)} \cdot t'(y) = \frac{a^{-1/6}}{\Gamma(r)} [t(y)]^a e^{-t(y)}, \quad x > -\frac{1}{b}$$

Choose the proposal density as $N(0, 1)$, then the envelope is

$$e(y) = \frac{1}{\sqrt{2\pi}} e^{-y^2/2} \cdot \frac{a^{-1/6}\sqrt{2\pi}}{\Gamma(r)} e^{a \log a - a} \geq f(y)$$

which holds when $y > -1/b$.

Sample X as follows.

- Sample $Z \sim N(0,1)$ and $U \sim \text{unif}(0,1)$
- Compute $X = t(Z) = a(1+bZ)^3$ and accept it if $X > 0$ and

$$\log(U) \leq \frac{1}{2}Z^2 + a\log\left(X/a\right) - X + a$$

The acceptance rate is high in general.

r	1	1.5	2	3	4
acceptance rate	0.951	0.973	0.982	0.989	> 0.99

Refer to Marsaglia and Tsang (2000). *ACM Transactions on Mathematical Software*, vol 26, no 3, pp 363–372.

Comments

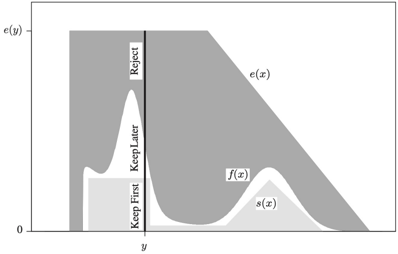
```
reject = function()
{
    x = propose(...)          # propose a value
    while( log(runif(1)) > log_f(x) - log_e(x) )
    {
        x = propose(...)      # propose another value
    }
    x
}
```

- It is enough to know $f(x)$ up to a proportional constant.
- Good rejection sampling envelopes have three properties: easy to construct, exceed the target everywhere, easy to sample
- Adaptive rejection sampling is proposed to handle the scenario when it is challenging to construct a suitable envelope.

Squeezed Rejection Sampling

If evaluating f is computationally expensive, we can apply squeezed rejection sampling. Let s be a suitable squeezing function, where $s(x)$ must not exceed $f(x)$ anywhere on the support of f.

- Sample $Y \sim g$
- Sample $U \sim \text{unif}(0, 1)$
- If $U \leq s(Y)/e(Y)$, output $X = Y$.
- Otherwise, if $U \leq f(Y)/e(Y)$, output $X = Y$.
- Otherwise, reject Y and repeat.



Example: Gamma Distribution

We want to avoid computing the two logarithms. Choose

$$s(y) = \frac{1}{\sqrt{2\pi}}(1 - 0.0331y^4)e^{-y^2/2} \cdot \frac{a^{-1/6}\sqrt{2\pi}}{\Gamma(r)}e^{a\log a - a}$$

The updated algorithm is as follows.

- Sample $Z \sim N(0,1)$ and $U \sim$ unif$(0,1)$
- Compute $X = t(Z) = a(1+bZ)^3$. If $X \leq 0$, resample Z
- Output X if $U \leq 1 - 0.0331Z^4$
- Output X if $\log(U) \leq Z^2/2 + a\log(X/a) - X + a$
- Otherwise, resample Z and U, then repeat

Sampling Importance Resampling Algorithm

Let $f(x)$ be the density and $g(x)$ be the density corresponding to a envelope for f. The sampling importance resampling (SIR) algorithm simulates realizations approximately from the target distribution f.

- Sample candidates $Y_1, \ldots, Y_m$ iid from g
- Calculate the weights $w(Y_1), \ldots, w(Y_m)$
- Resample $X_1, \ldots, X_n$ from $Y_1, \ldots, Y_m$ with replacement with probabilities $w(Y_1), \ldots, w(Y_m)$

We may view important sampling as approximating f by the discrete distribution having mass $w(x_i)$ on each observed point x_i for $i = 1, \ldots, m$.

Importance Weights

For the target density f, the weight used to correct sampling probabilities are the standardized importance weights

$$w(x_i) = \frac{f(x_i)/g(x_i)}{\sum_{i=1}^m f(x_i)/g(x_i)}$$

for a collection of values $x_1, \ldots, x_m$ drawn iid from an envelope g.

A random variable X drawn with the SIR algorithm has distribution that converges to f as $m \to \infty$. Let $w^*(x) = f(x)/g(x)$.

$$P(X \in \mathscr{A} \mid Y_1, \ldots, Y_m) = \frac{\sum_{i=1}^m 1(y_i \in \mathscr{A}) w^*(Y_i)}{\sum_{i=1}^m w^*(Y_i)} \to \frac{\int_{\mathscr{A}} w^*(y)g(y)dy}{\int w^*(y)g(y)dy}$$

$$P(X \in \mathscr{A}) = E[P(X \in \mathscr{A} \mid Y_1, \ldots, Y_m)] \to \int_{\mathscr{A}} f(y)dy$$

Comments

In principle, we require $n/m \to 0$ for distributional convergence of the sample. In practice, $10n \leq m$ works fine as long as the resulting resample does not contain too many replicates of any initial draw.

The SIR algorithm can be sensitive to the choice of g. The support of g must include the entire support of f. g should have heavier tails than f, or more generally, f/g never grows too large.

Difference between SIR and rejection sampling:

- Rejection sampling is exact, but requires a random number of draws to obtain a sample of size n
- SIR is approximate, and uses a pre-determined number of draws to generate a sample of size n

Example: Slash Distribution

Y has a slash distribution if $Y = X/U$, where $X \sim N(0,1)$ and $U \sim$ unif$(0,1)$ are independent. The slash density is

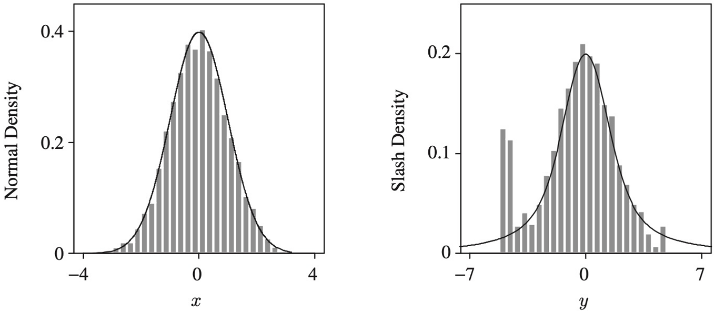
$$f(y) = \frac{1 - \exp(-y^2/2)}{y^2\sqrt{2\pi}}, \ y \neq 0, \quad f(y) = \frac{1}{2\sqrt{2\pi}}, \ y = 0.$$

Consider the following secanerios:

- Use the slash distribution as a SIR envelope to generate standard normal random variables
- Use the normal distribution as a SIR envelope to generate slash variates.

Results

Set $m = 100,000$ and $n = 5,000$.



The normal density is not a suitable importance sampling function for SIR use when generating draws from the slash distribution because the envelope's tails are far lighter than the target's.

Sequential Monte Carlo

SIR becomes inefficient when f is high dimensional. Sequential Monte Carlo methods address the problem by splitting the high-dimensional task into a sequence of simpler steps, each of which updates the previous one.

Let $X_{1:t} = (X_1, \ldots, X_t)$ be a discrete time stochastic process.

- X_t is the observation at time t
- $X_{1:t}$ is the entire history of the sequence so far
- f_t is the density of $X_{1:t}$

We want to estimate $E[h(X_{1:t})]$.

$$f_t(x_{1:t}) = f_1(x_1) f_2(x_2 \mid x_{1:1}) f_3(x_3 \mid x_{1:2}) \cdots f_t(x_t \mid x_{1:t-1})$$

Markov Processes

Assume that $X_{1:t}$ is a Markov Process.

$$f_k(x_k \mid x_{1:k-1}) = f_k(x_k \mid x_{k-1}), \quad k = 2, 3, \ldots, t.$$

The target density is

$$f_t(x_{1:t}) = f_1(x_1) f_2(x_2 \mid x_1) f_3(x_3 \mid x_2) \cdots f_t(x_t \mid x_{t-1})$$

Suppose that we adopt the same Markov form for the envelope,

$$g_t(x_{1:t}) = g_1(x_1) g_2(x_2 \mid x_1) g_3(x_3 \mid x_2) \cdots g_t(x_t \mid t_{t-1})$$

Given $x_{1:t-1}$ and w_{t-1}, we can take advantage of the Markov property by sampling only the next component, namely X_t, and then update the weight w_t immediately.

Sequential Importance Sampling

Draw $X_{1:t}$ from density $f_t(x_{1:t})$ with proposal density $g_t(x_{1:t})$.

1. Sample $X_1 \sim g_1$. Let $w_1 = u_1 = f_1(x_1)/g_1(x_1)$. Set $t = 2$
2. Sample $X_t \mid x_{t-1} \sim g_t(x_t \mid x_{t-1})$
3. Append x_t to $x_{1:t-1}$ to obtain $x_{1:t}$
4. Let $u_t = f_t(x_t \mid x_{t-1})/g_t(x_t \mid x_{t-1})$
5. Let $w_t = w_{t-1}u_t$. At the current time, w_t is the importance weight for $x_{1:t}$
6. Increment t and return to step 2.

Repeat the algorithm n times to get an iid sample $X_{1:t}^{(i)}$, $i = 1, \ldots, n$.

$$E[h(X_{1:t})] \approx \frac{\sum_{i=1}^n w_t^{(i)} h(X_{1:t}^{(i)})}{\sum_{i=1}^n w_t^{(i)}}$$

Example: Simple Markov Process

Let $X_t \mid X_{t-1} \sim f_t$ denote a Markov process, where

$$f_t(x_t \mid x_{t-1}) \propto |\cos(x_t - x_{t-1})| \exp\left\{-\frac{1}{4}(X_t - X_{t-1})^2\right\}$$

Suppose we wish to estimate σ_t, the standard deviation of X_t. Let the sample be $X_{1:t}^{(i)}$ with importance weights $w_t^{(i)}$ for $i = 1, \ldots, n$.

$$\hat{\sigma}_t = \left[\frac{1}{1 - \sum_i (\tilde{w}_t^{(i)})^2} \sum_{i=1}^{n} \tilde{w}_t^{(i)} (x_t^{(i)} - \hat{\mu}_t)^2\right]^{1/2}$$

where $\tilde{w}_t^{(i)} = w_t^{(i)} / \sum_i w_t^{(i)}$ and $\hat{\mu}_t = \sum_i \tilde{w}_t^{(i)} x_t^{(i)}$.

At stage t, we sample from $X_t^{(i)} \mid x_{t-1}^{(i)} \sim N(x_{t-1}^{(i)}, 1.5^2)$. The weight updates are

$$u_t^{(i)} \propto \frac{|\cos(x_t^{(i)} - x_{t-1}^{(i)})| \exp\left\{-(x_t^{(i)} - x_{t-1}^{(i)})^2/4\right\}}{\phi(x_t^{(i)}; x_{t-1}^{(i)}, 1.5^2)}$$

At $t = 100$, we find $\hat{\sigma}_t = 13.4$.

As t increases, the weights develop an undesirable degeneracy, that is, the weights become concentrated on only a few sequences $X_{1:t}$, which degrades estimation performance.

Coefficient of Variation

Degeneracy of the weights is related to their variability.

- More severe degeneracy corresponds to the larger variance of the weights.

A measure of variability is the squared coefficient of variation,

$$\mathrm{CV}^2[w(X)] = \frac{var[w(X)]}{(E[w(X)])^2} = \frac{E(w(X) - n^{-1})^2}{n^{-2}} = E[nw(X) - 1]^2$$

where $w(X)$ is the standardized importance weight. Estimate it by

$$\widehat{\mathrm{CV}}^2[w(X)] = \frac{1}{n}\sum_{i=1}^{n}\left(nw(x^{(i)}) - 1\right)^2 = n\sum_{i=1}^{n} w(x^{(i)})^2 - 1$$

Effective Sample Size

The effective sample size (ESS) can be used to measure the efficiency of using an envelope g with target f.

- The ESS can be interpreted as indicating that the n weighted samples used in an importance sampling estimate are worth a certain number of unweighted iid samples drawn exactly from f.

We can measure the effective sample size as

$$\hat{N}(g, f) = \frac{n}{1 + \widehat{\mathrm{CV}}^2[w(X)]} = \frac{1}{\sum_{i=1}^n w(x^{(i)})^2}$$

Larger effective sample sizes are better.

Example for Illustration

Consider a scenario

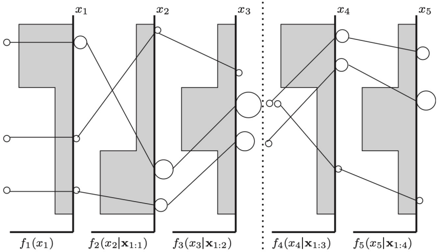
- n samples have normalized importance weights $w(x^{(i)})$
- z of these weights are zero
- remaining $n-z$ of these weights are $1/(n-z)$

Simple calculation suggests,

$$\widehat{\mathsf{CV}}^2[w(X)] = \frac{n}{n-z} - 1, \qquad \hat{N}(g,f) = n - z$$

Rejuvenation

- For sequential importance sampling, monitor $\hat{N}_t(g, f)$ to assess importance weight degeneracy.
- The ESS will decrease as time progresses.
- When the ESS falls below some threshold, rejuvenate the collection of sequences $X_{1:t-1}^{(i)}$, $i = 1, \ldots, n$.
    - Resample the sequences with replacement with probabilities equal to $w_t(x_{1:t}^{(i)})$ and then reset all weights to $1/n$.



General Sequential Importance Sampling

When the Markov property does not apply,

$$f_t(x_{1:t}) = f_1(x_1) f_2(x_2 \mid x_{1:1}) f_3(x_3 \mid x_{1:2}) \cdots f_t(x_t \mid x_{1:t-1})$$

The envelope is

$$g_t(x_{1:t}) = g_1(x_1) g_2(x_2 \mid x_{1:1}) g_3(x_3 \mid x_{1:2}) \cdots g_t(x_t \mid x_{1:t-1})$$

The importance weight is

$$\begin{aligned} w_t(x_{1:t}) &= \frac{f_1(x_1) f_2(x_2 \mid x_{1:1}) f_3(x_3 \mid x_{1:2}) \cdots f_t(x_t \mid x_{1:t-1})}{g_1(x_1) g_2(x_2 \mid x_{1:1}) g_3(x_3 \mid x_{1:2}) \cdots g_t(x_t \mid x_{1:t-1})} \\ &= w_{t-1}(x_{1:t-1}) \frac{f_t(x_t \mid x_{1:t-1})}{g_t(x_t \mid x_{1:t-1})} \end{aligned}$$

Example: High Dimensional Distribution

Consider a sequence of probability densities given by

$$f_t(x_{1:t}) = k_t \exp\left\{-\|x_{1:t}\|^3/3\right\}$$

for some constants k_t.

1. Let $t = 1$. Sample n points $X_1^{(1)}, \ldots, X_1^{(n)}$ iid from $N(0,1)$. Calculate the initial weights as $w_1^{(i)} = \exp\{-|x_1^{(i)}|^{3/2}\}/\phi(x_1^{(i)})$
2. For $t > 1$, sample n points $X_t^{(i)}$ from $N(0,1)$. Append these to $X_{1:t-1}^{(i)}$, yielding $X_{1:t}^{(i)}$.
3. Calculate $u_t^{(i)} = f(x_t^{(i)} \mid x_{1:t-1}^{(i)})/\phi(x_t^{(i)})$
4. Set $w_t^{(i)} = w_{t-1}^{(i)} u_t^{(i)}$ and standardize the new weights
5. Calculate $\hat{N}_t(\phi, f_t)$. If $\hat{N}_t(\phi, f_t) < \alpha n$ then resample the current sample of $X_{1:t}^{(i)}$ with replacement with probabilities equal to the $w_t^{(i)}$. Discard the original sample and reset all weights to $1/n$.
6. Increment t and return to step 2 until $t = p$.

Comments

- The resampling strategy in step 5 attempts to rejuvenate the sample periodically to reduce degeneracy; hence the algorithm is called sequential importance sampling with resampling.
- Let α control the tolerance for degeneracy.
- As an example, set $p = 50$, $n = 5000$, and $\alpha = 0.2$.
  - Over the 50 stages, step 5 was trigered 8 times.
  - The median ESS was about 2000.
- As a comparison, draw 5000 points from a 50-dimensional standard normal envelope and importance reweight these points all at once. The ESS is about 1.13.

Variance Reduction Techniques

Recall that the Monte Carlo estimator of $E[h(X)]$ is

$$\hat{\mu}_{\text{MC}} = \frac{1}{n}\sum_{i=1}^{n} h(X_i) \quad \longrightarrow \quad E[h(X)] = \int f(x)f(x)dx$$

where $X_1, \ldots, X_n$ are randomly sampled from f.

$$var(\hat{\mu}_{\text{MC}}) = \frac{1}{n} var[h(X)], \quad \text{if } X_i \text{ are iid}$$

We can reduce the variance of the estimator by

- importance sampling
- antithetic sampling
- control variates
- Rao-Blackwellization

Importance Sampling: Motivation

Let X be the result of rolling a die. Estimate $p = P(X = 1)$.

- Use a balanced die. Roll the die n times, and estimate p by the proportion of $X_i = 1$.
- Use a modified die, where 2 and 3 are also marked as 1. Roll the die n times and estimate p by $1/3$ of the proportion of $X_i = 1$.

$$var(\hat{p}_1) = \frac{5}{36n}, \quad var(\hat{p}_2) = \frac{1}{36n}$$

The first approach needs a much larger sample size to achieve the same variance than the second approach.

An importance sampling distribution is used to oversample a portion of the state space that receives lower probability under the target distribution. An importance weighting corrects for this bias and can provide an improved estimator.

Importance Sampling: Rational

Notice that

$$\mu = \int h(x)f(x)dx = \int h(x)\frac{f(x)}{g(x)}g(x)dx$$

where g is another density function, called the importance sampling function or envelope. We can estimate $E[h(X)]$ by drawing $X_1, \ldots, X_n$ iid from g and

$$\hat{\mu}^*_{\mathsf{IS}} = \frac{1}{n}\sum_{i=1}^{n} h(X_i)w^*(x_i)$$

where $w^*(x_i) = f(x_i)/g(x_i)$ are unstandardized weights, also called importance ratios.

Standardized Weights

Notice that

$$\mu = \frac{\int h(x) f(x) dx}{\int f(x) dx} = \frac{\int h(x)[f(x)/g(x)]g(x)dx}{\int [f(x)/g(x)]g(x)dx}$$

Hence, we can also estimate

$$\hat{\mu}_{\mathsf{IS}} = \frac{1}{n} \sum_{i=1}^{n} h(X_i) w(x_i)$$

where $w(x_i) = w^*(x_i)/\sum w^*(x_i)$ are standardized weights. This is particularly important when f is known only up to a proportionality constant.

Comments

- $f(x)/g(x)$ should be bounded and g has heavier tails than f. Otherwise, some standardized importance weights will be huge.
- The ESS can be used to measure the efficiency of an importance sampling strategy using envelope g.
- The choice between using the unstandardized and the standardized weights depends on several considerations.
  - $\hat{\mu}^*_{\mathsf{IS}}$ is unbiased, while $\hat{\mu}_{\mathsf{IS}}$ is biased
  - $\hat{\mu}_{\mathsf{IS}}$ has smaller MSE when $w^*(x)$ and $h(x)w^*(x)$ are strongly correlated.
  - An advantage of using the standardized weights is that is does not require knowing the proportionality constant for f.

Comparing Importance Sampling with SIR

Suppose that an initial sample $Y_1, \ldots, Y_m$ with corresponding weights $w(Y_1), \ldots, w(Y_m)$ is resampled to provide n SIR draws $X_1, \ldots, X_n$, where $n < m$. Let

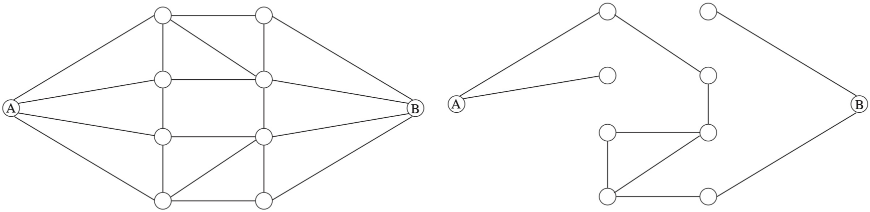
$$\hat{\mu}_{\mathsf{SIR}} = \frac{1}{n} \sum_{i=1}^{n} h(X_i)$$

The SIR estimator has the same bias as $\hat{\mu}_{\mathsf{IS}}$. However,

$$var(\hat{\mu}_{\mathsf{SIR}}) \geq var(\hat{\mu}_{\mathsf{IS}})$$

Example: Network Failure Probability

A signal sent from A to B must follow a path along available edges. Want to compute the probability of network failure given specific probabilities for the failure of each edge.



Assume each edge may fail independently with the same probability p. Let $X = (X_1, \ldots, X_{20})$ be a network with 20 edges. Let $b(X)$ be the number of broken edges in X. Let $h(X) \in \{0, 1\}$ be the status of network, where 1 for not connected and 0 for connected.

Example: Continued

- Naive Monte Carlo: draw $X_1, \ldots, X_n$ independently and uniformly from the set of all possible network configures, and

$$\hat{\mu}_{\mathsf{MC}} = \frac{1}{n} \sum_i h(X_i), \qquad var(\hat{\mu}_{\mathsf{MC}}) = \frac{\mu - \mu^2}{n}$$

- Importance sampling: Simulate $X_1^*, \ldots, X_n^*$ by generating network configures formed by breaking edges, assuming independent edge failure with probability $p^* > p$. The importance weight

$$w^*(X_i^*) = \left(\frac{1-p}{1-p^*}\right)^{20} \left[\frac{p(1-p^*)}{p^*(1-p)}\right]^{b(X_i^*)}$$

$$\hat{\mu}_{\mathsf{IS}}^* = \frac{1}{n} \sum_{i=1}^{n} h(X_i^*) w^*(X_i^*)$$

Example: Continued

Let $p = 0.05$ and $p^* = 0.25$. We find $w^*(X) \leq 0.07$.

$$var(\hat{\mu}^*_{\mathsf{IS}}) \leq \frac{0.07\mu - \mu^2}{n} < var(\hat{\mu}_{\mathsf{MC}})$$

Hence, $var(\hat{\mu}_{\mathsf{MC}})/var(\hat{\mu}^*_{\mathsf{IS}}) \approx 14$.

Simulation results based on $n = 100,000$ samples:

	naive Monte Carlo	importance sampling
# of failures	2	491
$\hat{\mu}$	2×10^{-5}	1.02×10^{-5}
SE($\hat{\mu}$)	1.41×10^{-5}	1.57×10^{-6}

Antithetic Sampling: Motivation

Let $\hat{\mu}_1$ and $\hat{\mu}_2$ be two identically distributed unbiased estimators and are negatively correlated.

$$\hat{\mu}_{\mathsf{AS}} = \frac{\hat{\mu}_1 + \hat{\mu}_2}{2}, \qquad var(\hat{\mu}_{\mathsf{AS}}) = \frac{(1+\rho)\sigma^2}{2n} < var(\hat{\mu}_1) = \frac{\sigma^2}{n}$$

where ρ is the correlation between $\hat{\mu}_1$ and $\hat{\mu}_2$ and σ^2 is the variance of $\hat{\mu}_1$ or $\hat{\mu}_2$.

Such pairs of estimators can be generated using the antithetic sampling approach. Generate

Antithetic Sampling: Rationale

If h_1 and h_2 are functions of m random variables $U_1, \ldots, U_m$ and if each function is monotone in each argument, then

$$cov[h_1(U_1, \ldots, U_m), h_2(1 - U_1, \ldots, 1 - U_m)] \leq 0$$

Let $X = \{X_1, \ldots, X_n\}$ denote a set of iid random variables and $h(X_i) = h(X_{i1}, \ldots, X_{im})$ be monotone in its arguments.

$$\hat{\mu}_1(X) = \frac{1}{n} \sum_{i=1}^{n} h(F_1^{-1}(U_{i1}), \ldots, F_m^{-1}(U_{im}))$$

$$\hat{\mu}_2(X) = \frac{1}{n} \sum_{i=1}^{n} h(F_1^{-1}(1 - U_{i1}), \ldots, F_m^{-1}(1 - U_{im}))$$

where F_j is the CDF of each X_{ij} and U_{ij} are iid unif$(0, 1)$. Then the estimator is $\hat{\mu}_{\text{AS}} = (\hat{\mu}_1 + \hat{\mu}_2)/2$.

Example: Normal Expectation

Let X be $N(0,1)$ and $h(x) = x/(2^x - 1)$. Compute $\mu = E[h(X)]$.

- Naive Monte Carlo: Sample $X_1, \ldots, X_n \sim N(0,1)$

$$\hat{\mu}_{\text{MC}} = \frac{1}{n}\sum_{i=1}^{n} h(X_i), \quad \hat{\mu}_{\text{AS}} = \frac{1}{n}\sum_{i=1}^{n/2}(h(X_i) + h(-X_i))$$

- Antithetic sampling: Sample $X_1, \ldots, X_{n/2} \sim N(0,1)$.

Notice that $cor[h(X), h(-X)] = -0.95$ and based on $n = 100,000$,

$$\hat{\mu}_{\text{MC}} = 1.4993, \quad \text{SE}(\hat{\mu}_{\text{MC}}) = 0.0016$$
$$\hat{\mu}_{\text{AS}} = 1.4992, \quad \text{SE}(\hat{\mu}_{\text{AS}}) = 0.0003$$

Example: Network Failure Probability

Each simulated network X_i is determined by $U(0,1)$ random variables $U_{i1}, \ldots, U_{im}$, where $m = 20$.

- The jth edge in X_i is broken if $U_{ij} < p$
- The antithetic network X_i^* is obtained by breaking the jth edge when $U_{ij} > 1 - p$

and estimate the failure probability as

$$\frac{1}{2n} \sum_{i=1}^{n} (h(X_i) + h(X_i^*))$$

Note that the same $U_{i1}, \ldots, U_{im}$ yield both X_i and X_i^*.

Control Variates

The control variate strategy improves estimation of an unknown integral by relating the estimate to some correlated estimator of an integral whose value is known.

Want to estimate $\mu = E[h(X)]$ and we know $\theta = E[c(Y)]$. Let $(X_1, Y_1), \ldots, (X_n, Y_n)$ denote pairs of random variables observed independently.

$$cov(X_i, X_j) = cov(Y_i, Y_j) = cov(X_i, Y_j) = 0, \quad i \neq j.$$

The simple Monte Carlo estimator is

$$\hat{\mu}_{\mathsf{MC}} = \frac{1}{n} \sum_i h(X_i), \quad \hat{\theta}_{\mathsf{MC}} = \frac{1}{n} \sum_i c(Y_i)$$

The control variate estimator

$$\hat{\mu}_{\mathsf{CV}} = \hat{\mu}_{\mathsf{MC}} + \lambda(\hat{\theta}_{\mathsf{MC}} - \theta)$$

Choose λ to minimize the $var(\hat{\mu}_{\mathsf{CV}})$.

$$\lambda = \frac{-cov(\hat{\mu}_{\mathsf{MC}}, \hat{\theta}_{\mathsf{MC}})}{var(\hat{\theta}_{\mathsf{MC}})}$$

where the unknown moments of $h(X_i)$ and $c(Y_i)$ can be estimated using the samples,

$$\widehat{var}(\hat{\theta}_{\mathsf{MC}}) = \frac{1}{n(n-1)} \sum_{i=1}^{n} [c(Y_i) - \bar{c}]^2,$$

$$\widehat{cov}(\hat{\mu}_{\mathsf{MC}}, \hat{\theta}_{\mathsf{MC}}) = \frac{1}{n(n-1)} \sum_{i=1}^{n} [h(X_i) - \bar{h}][c(Y_i) - \bar{c}]$$

Comments

- The minimum variance is

$$\min_{\lambda} var(\hat{\mu}_{\mathsf{CV}}) = var(\hat{\mu}_{\mathsf{MC}}) - \frac{(cov(\hat{\mu}_{\mathsf{MC}}, \hat{\theta}_{\mathsf{MC}}))^2}{var(\hat{\theta}_{\mathsf{MC}})}$$

- Consider regression $E[h(X_i) \mid Y_i = y_i] = \beta_0 + \beta_1 c(y_i)$
  - $\hat{\lambda} = -\hat{\beta}_1$ negative slope
  - $\hat{\mu}_{\mathsf{CV}} = \hat{\mu}_{\mathsf{MC}} + \lambda(\hat{\theta}_{\mathsf{MC}} - \theta) = \hat{\beta}_0 + \hat{\beta}_1\theta$ fitted response at θ
  - SE($\hat{\mu}_{\mathsf{CV}}$) is the SE of the fitted response
- In practice, $\hat{\mu}_{\mathsf{MC}}$ and $\hat{\theta}_{\mathsf{MC}}$ often depend on the same random variables, $X_i = Y_i$.
- It is possible to use more than one control variate.

$$\hat{\mu}_{\mathsf{CV}} = \hat{\mu}_{\mathsf{MC}} + \sum_{j=1}^{m} \lambda_j(\hat{\theta}_{\mathsf{MC},j} - \theta_j)$$

Control Variate for Importance Sampling

Let $w^*(x)$ be the weights in importance sampling. Because $E[w^*(X)] = 1$, the importance sampling control variate estimator is

$$\hat{\mu}_{\mathsf{ISCV}} = \hat{\mu}^*_{\mathsf{IS}} + \lambda(\bar{w}^* - 1)$$

where $\bar{w}^* = \sum_i w^*(x_i)/n$.

Fit a linear regression with response being $h(x_i)w^*(x_i)$ and predictor being $w^*(x_i)$.

- $\hat{\lambda}$ is the negative slope
- $\hat{\mu}_{\mathsf{ISCV}}$ is the fitted response at 1.
- $\mathsf{SE}(\hat{\mu}_{\mathsf{ISCV}})$ is the SE of the fitted response at 1.

Rao-Blackwellization

Suppose that each $X_i = (X_{i1}, X_{i2})$ and the conditional expectation $E[h(X_{i1}) \mid X_{i2}]$ can be solved for analytically. The Rao-Blackwellized estimator is

$$\hat{\mu}_{\text{RB}} = \frac{1}{n} \sum_{i=1}^{n} E[h(X_i) \mid X_{i2}]$$

Notice that

$$var(\hat{\mu}_{\text{MC}}) \geq var(\hat{\mu}_{\text{RB}})$$

Rao-Blackwellization of Rejection Sampling

For rejection sampling

- Draw candidates $Y_1, \ldots, Y_M$ and some are rejected
- Uniform random variables $U_1, \ldots, U_M$
- Reject Y_i if $U_i > w^*(Y_i)$, where $w^*(Y_i) = f(Y_i)/e(Y_i)$
- Stop at a random time M after accepting n draws

The estimators of $\mu = E[h(X)]$ are

$$\hat{\mu}_{\text{MC}} = \frac{1}{n}\sum_{i=1}^{n} h(Y_i) I_{[U_i \le w^*(Y_i)]}, \quad \hat{\mu}_{\text{RB}} = \frac{1}{n}\sum_{i=1}^{n} h(Y_i) t_i(Y),$$

where $t_i(Y) = P[U_i \le w^*(Y_i) \mid M, Y_1, \ldots, Y_M]$.